AWANTIKA SRIVASTAVA

AI/ML Engineer | Machine Learning Systems | Performance Optimization

+91-8920482037 | sawantika81@gmail.com | LINKEDIN | Github

PROFILE SUMMARY

AI/ML Engineer with 2+ years of experience in machine learning model development and performance optimization. Strong foundation in data structures, algorithms, object-oriented programming, and system-level understanding of ML frameworks. Experienced in optimizing inference workflows, understanding compute-memory trade-offs, and working with ONNX and runtime-based deployment pipelines.

CORE TECHNICAL SKILLS

- **Programming & Data Science:** Python (OOP concepts), C++, SQL., Pandas, Numpy, Scikit-learn, Object-Oriented Design
- **Statistics & Mathematics:** Statistical Modeling, Descriptive Statistics, Hypothesis Testing, Probability, Sampling, A/B Testing, Cross-Validation
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, LightGBM, Model Evaluation Metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC).
- **Deep Learning:** ANN, CNN, RNN, LSTM, BERT, Transfer Learning, Self-Attention, multi-head Attention, Time-Series Forecasting, ARIMA, SARIMA
- **NLP:** Text preprocessing, Tokenization, Chunking, Sentiment Analysis, Transformer-based models, NLTK, TextBlob, SpaCy
- **ML Frameworks:** PyTorch, TensorFlow, ONNX Runtime
- **Computer Vision:** Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Paddle OCR, Video Analytics (RTSP), OpenCV
- **Model Optimization:** Inference Optimization, Quantization (INT8/FP16 – Conceptual), ONNX Conversion
- **ML Operations Awareness:** GEMM, Convolution (CONV), Matrix Multiplication Basics
- **Systems Understanding:** CPU/GPU Compute Awareness, Memory Utilization Concepts, Performance Benchmarking
- **Tools:** Linux, Git, GitHub, Docker

EXPERIENCE

AI/ML Engineer | PPS International Pvt. Ltd. **January 2024-Present**

- Developed and optimized **machine learning models** for **real-time inference** with focus on latency and throughput improvements.
- Converted trained models to **ONNX format** and evaluated **ONNX Runtime performance** across different execution backends.
- Conducted detailed **performance benchmarking** and profiling to identify computational bottlenecks in inference pipelines.
- Analyzed cost of key ML operations such as **GEMM (matrix multiplication)** and **Convolution (CONV)** to understand optimization opportunities.
- Applied **model optimization techniques** including graph simplification and precision-aware inference (FP16/INT8 – conceptual) to reduce memory and compute overhead.
- Evaluated **compute–memory trade-offs** and improved resource utilization in CPU-based environments.
- Implemented **object-oriented programming (OOP)** principles and followed **software engineering best practices** to build modular and maintainable systems.
- Worked in **Linux environments**, leveraging **Git-based version control** and collaborative workflows to ensure high-quality and scalable code development.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- **Designed and deployed a** real-time computer vision–based ML system **for unsafe** driver behavior detection using **CNN-based SSD MobileNet** model.
- Trained and optimized models on large-scale video datasets, achieving **20–25 FPS** real-time processing with **<150 ms inference latency**.
- Implemented **end-to-end ML pipelines** for data ingestion, preprocessing, model training, evaluation, and production inference.
- Deployed optimized models using **TensorFlow Lite** on edge/production environments for continuous monitoring.
- Built a **Flask-based web dashboard** to visualize detections and automatically record **30-second event clips**, reducing manual review effort.

Chatbot Using LLM & RAG | Applied ML Project

- Built a Lightweight **LLM-Powered chatbot** using **TinyLLaMA** to answer user queries over content.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant resume sections for contextual question answering.
- Selected **TinyLLaMA** to ensure **low memory footprint and fast inference**, making the solution suitable for resource-constrained environments.
- Applied **prompt engineering techniques** to improve response relevance and consistency.
- Deployed the chatbot as an interactive **Streamlit web application** for real-time user interaction.

YouTube Comments Sentiment Analyzer | link-https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Developed **LSTM-based time-series forecasting** model trained on 5+ years historical stock data reducing **RMSE** by 18**%.**
- Engineered lag **features**, **rolling statistics**, and volatility indicators improving prediction stability.
- **Implemented** custom **BPTT training** loop with adaptive learning rate scheduling for **stable convergence**.
- Applied **regularization techniques** reducing overfitting gap by **20**%.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**

Bachelor of Technology (Electrical and electronics engineering)